

Day 18 - Lecture 18.16: Multiple Datasets on Scatter Plots

1. Overview & Purpose

Comparing multiple clusters or seasonal datasets on a single scatter plot reveals group clustering, separation boundaries, and seasonal shifts.

2. Case Study: Weather Conditions (Summer vs. Winter)

Analyzing temperature ($^{\circ}\text{C}$) vs humidity (%) across 5 cities during two distinct seasons:

- **Winter:** Low temperature ($0^{\circ}\text{C} - 10^{\circ}\text{C}$), High humidity (65) $\rightarrow$ Clustered at top-left.
- **Summer:** High temperature ($25^{\circ}\text{C} - 35^{\circ}\text{C}$), Moderate humidity (45) $\rightarrow$ Clustered at bottom-right.



3. Code Implementation

```
import matplotlib.pyplot as plt

cities = ["City A", "City B", "City C", "City D", "City E"]

# Winter: Temperature (°C) vs Humidity (%)
winter_temp = [5, 2, 10, 0, 7]
winter_humidity = [80, 75, 65, 85, 70]

# Summer: Temperature (°C) vs Humidity (%)
summer_temp = [25, 30, 28, 35, 27]
summer_humidity = [60, 50, 55, 45, 65]

plt.figure(figsize=(9, 6))

# Plot seasonal clusters
plt.scatter(winter_temp, winter_humidity, color="royalblue", s=90, label="Winter")
plt.scatter(summer_temp, summer_humidity, color="darkorange", s=90, label="Summer")

plt.title("Seasonal Weather Comparison: Temperature vs Humidity", fontsize=14, fontweight="bold")
plt.xlabel("Temperature (°C)", fontsize=12)
plt.ylabel("Humidity (%)", fontsize=12)
plt.legend(loc="upper right", fontsize=11)
plt.grid(True, linestyle="--", alpha=0.6)
plt.show()
```

4. Key Takeaways

- Using distinct markers (circles for Winter, squares for Summer) along with distinct colors makes the visualization accessible to colorblind readers.
- Scatter plots instantly expose whether two classes are linearly separable for classification models (e.g. SVM or Logistic Regression).